INTERNATIONAL MONETARY FUND

MALI

Debt Sustainability Analysis

Prepared by a staff team from the Afican and
Policy Development and Review Departments, and the World Bank

Approved by Benedicte Vibe Christensen and Matthew Fisher

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External Debt Sustainability Assessment

1. Since the 1994 CFA franc devaluation, which increased debt and debt service indicators substantially, Mali has benefited from a series of debt relief measures that have reduced debt service, including the Highly Indebted Poor Country (HIPC) Initiative. The HIPC Initiative reduced debt service to sustainable levels as defined by a specific threshold for the NPV of debt to exports (200 percent original HIPC Initiative and 150 percent for the enhanced HIPC Initiative). Despite the implementation of sound macroeconomic policies and prudent debt management, the capacity to repay debt remains subject to moderate risks from natural and commodity price shocks. Accordingly, lessening the exposure to such shocks through diversification and improving productivity remains key to lessening medium-term debt vulnerabilities.

2. This DSA applies the low income country template to the staff's baseline projections, and applies various robustness checks.[1] The analysis shows that external debt sustainability is broadly achieved under baseline projections in relation to the policy dependent debt burden thresholds. However, adverse external and internal shocks could result in breaching debt-to-exports and debt-to-GDP thresholds over the medium term. Accordingly Mali is assessed as a having a moderate risk of debt distress. Accelerating export growth and maintaining highly concessional loan assistance are important factors in maintaining debt

[1] The low income debt sustainability framework compares debt burden indicators to indicative policy-based thresholds. The thresholds are based on the empirical finding that low-income countries with stronger policies and institutions tend to have a higher debt carrying capacity. See IDA and IMF, "Operational Framework for Debt Sustainability Assessments in Low-Income Countries—Further Considerations" (IDA/R2005-0056), April 2005.

sustainability. The multilateral debt reduction initiative (MDRI) expected to be implemented in 2006 is also projected to improve the debt ratios significantly over the medium term.

A. Baseline projections

3. The DSA focuses on the external debt situation, since the existing domestic debt stock is negligible (less than 2 percent of GDP) while projected budgetary financing needs are likely to be covered by external grant and concessional loan assistance for the foreseeable future.

Mali. Stock of Domestic Debt, 2002-2005

	2002	2003	2004	2005 September
	(billions of CFAF unless otherwise noted)			
Gross debt to the banking sector	49.8	44.9	34.0	28.7
Government Securities	18.3	8.0	7.5	7.5
Non-bank debt	39.0	25.3	12.4	17.5
Total domestic public debt	107.1	78.2	53.9	53.7
Total domestic public debt (percent of GDP)	4.6	3.0	2.1	1.9

Source: Malian authorities.

4. Mali's external debt ratios have improved compared to the projections made at the completion point (2003) mainly reflecting a CFA franc appreciation against the U.S. dollar that has lowered the ratio of external debt to GDP, and better-than-expected US dollar-denominated exports in 2004 that has improved the debt service ratio and the NPV of debt-to-exports ratio. The NPV of debt-to-GDP in 2004 has dropped from 33 percent at the completion point to 25 percent in the DSA. The NPV of debt-to-exports in 2005 is reduced from 118 percent to 106 percent while the debt service ratio is unchanged at 6 percent in both analyses.

5. Macroeconomic projections are based on 2004 actual data, 2005 estimates for debt and the principal macroeconomic indicators. Projections for key macroeconomic variables have not been substantively changed compared to enhanced HIPC Initiative completion point projections made in March 2003. Under the baseline scenario, projected GDP growth rises gradually compared to the ten-year average of 5.5 percent (1995-2004), from 5.6 percent during 2005-10 to 6 percent during 2011-25. The improvement in growth reflects an assumption that a gradually increasing role of the private sector leads to more efficient investment and production decisions, as well as gradual improvements in the quality of human capital. The key sector developments incorporated in this scenario are: (i) broad-based production growth—agriculture (reflecting both increased planted area and yield),

cotton processing, energy, trade, and services; (ii) a projected decline in gold output after 2008, followed by an assumed leveling off of production from 2014 onward (at 50 tons per year equivalent to 2005 estimated production); and (iii) increased growth in nonmining exports over the longer term, helping to offset the substantial slowdown in export growth in 2010 and subsequent years as a result of the fall in gold production (Box 1). This improvement results from diversification engendered by removing impediments to trade and investment at the country and sub-regional level, and a gradual improvement of financial intermediation by the banking system.

Box 1. Mali: Debt Sustainability Analysis: Macroeconomic Assumptions, 2005-25

- Real GDP is projected to grow by 5.6 percent per annum 2005-10 and 6.0 percent on average during 2011-25.
- Inflation remains at about 2.5 percent during the projection period.
- Government revenue is stable at about 19 percent of GDP.
- The fiscal deficit, excluding grants, declines to 6.3 percent of GDP in 2010 and to less than 6 percent thereafter. Including grants, the deficit declines to less than 3.7 percent of GDP.
- The noninterest current account deficit averages 8.0 percent of GDP during 2005-10 then weakens to 9.3 percent of GDP during 2011-25.
- The volume of exports grows by about 9 percent during 2005-10; given the expected decline in gold exports, it averages 4 percent thereafter. The terms of trade and real exchange rate are projected to remain unchanged from 2010 onwards.
- The volume of imports is projected to grow in line with real GDP.
- Public sector external borrowing averages 4-5 percent of GDP a year; loans are assumed to have an average maturity of 28 years, a grace period of 7 years, an average interest rate of 0.7 percent, and a grant element of 50 percent. It is assumed that 65 percent of the new borrowings will come from multilateral sources and the remaining 35 percent from bilateral sources. IDA disbursements are assumed to be loans.
- Net private capital inflows are assumed to average 2½ percent of GDP a year (compared to 2.3 percent over the period 1995-2003).
- A uniform discount rate of 5 percent is used for NPV calculations.

6. **Mali's external debt ratios are projected to remain below, or close to, applicable debt thresholds over the period 2005-25** (Box 2, Figure 1, Table 1). Based on the 2004 World Bank Country Policy Institutional Assessment (CPIA) ratings, Mali is classified as a medium performer, which determines the relevant debt thresholds. In particular:

- the NPV of debt-to-GDP ratio is projected to fall from an average of 28 percent during 2005-10 to 22 percent in 2025, as a result of steady GDP growth;

- the NPV of the debt-to-exports ratio increases over the projection period, particularly after 2010 as export growth slows, and breaches the threshold level of 150 percent by a moderate amount at the end of the projection period; and,

- the debt service-to-export ratio is projected to decline marginally during the period 2005-2010, and is projected to stabilize from 2010 onwards;

Box 2: Policy-Based External Debt Burden Indicators

	Thresholds[1/]	Mali: Baseline Scenario Ratios		
		2004	2005-10[2/]	2011-25[2/]
		(In percent)		
NPV of debt-to-exports	150	106	100	136
NPV of debt-to-GDP	40	25	28	28
Debt service-to-exports	20	6.4	6.5	5.5

1/ Shows policy indicative thresholds as used in the joint IMF-World Bank low-income country DSA framework for a medium policy performer.
2/ Simple average.

B. Standard and bound tests for baseline

7. **The standard set of sensitivity checks on the evolution of debt ratios suggest that Mali continues to be vulnerable to external shocks over the period 2005–25** (Figure 2, Table 2). Under an alternative scenario which sets key parameters to their historical average values, debt to GDP and the NPV of debt to exports rise above the relevant threshold levels by 2010, although owing to a high degree of concessional loan assistance, the debt service ratio remains below the threshold. The key factor in the historical scenario is the high noninterest current account deficit (14.3 percent of GDP compared to projections of around 9 percent of GDP). In a second scenario, with public sector loans at less favorable terms, the external debt stock ratio indicates that Mali becomes at risk of debt distress by the end of the projection period, underscoring the importance of retaining high levels of concessional assistance (Tables 1 and 2, Figure 1).

8. **The *bound* sensitivity tests, assume adverse shocks to key parameters first separately, and in some cases jointly**. The main results of these sensitivity tests are that despite improved GDP growth and containment of inflation, Mali remains vulnerable to exogenous shocks, particularly:

- a one-time 30 percent nominal depreciation relative to the baseline would result in a sharp increase in the NPV of debt-to-GDP ratio in 2006, which would cross the threshold by 2007 placing Mali at risk of debt distress until 2022; and,

- a temporary shock on export growth of one standard deviation is significant in magnitude owing to the high standard deviation of exports. The NPV of debt-to-exports ratio reaches 153 percent in 2007.

9. **Based on these findings, this analysis suggests that Mali has a moderate risk of debt distress**. The baseline scenario indicates that the debt-to-exports ratio is marginally above the debt burden threshold toward the end of the projection period. Stress testing results in the debt-to-GDP and debt-to-exports indicators exceeding thresholds over the medium term (for export shocks, a 30 percent devaluation scenario and using historical averages of key data), though the debt service-to-exports indicator remains well below the 20 percent threshold in all cases. The low level of the debt service-to-exports indicator in both the baseline and the stress testing highlights the importance of maintaining a high degree of concessionality in future lending to Mali. Risks would clearly increase if, over the medium term, the level of concessionality were to decrease.

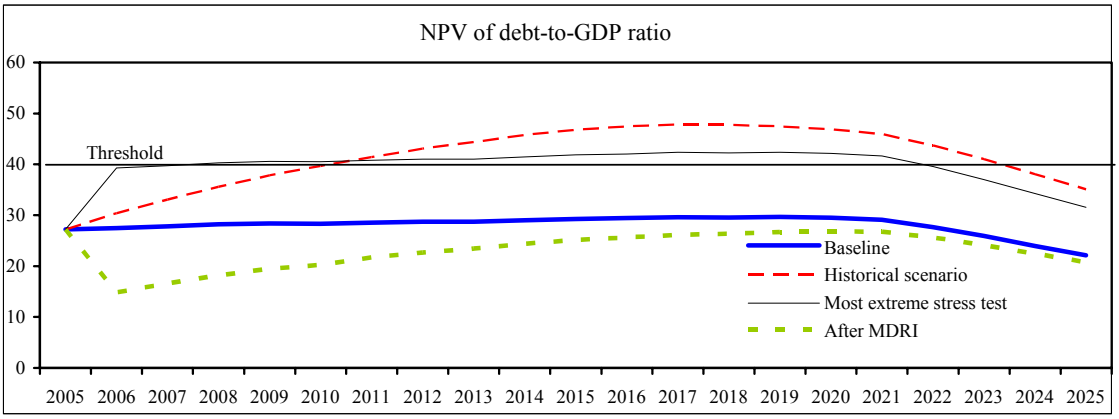
C. Multilateral Debt Relief Initiative

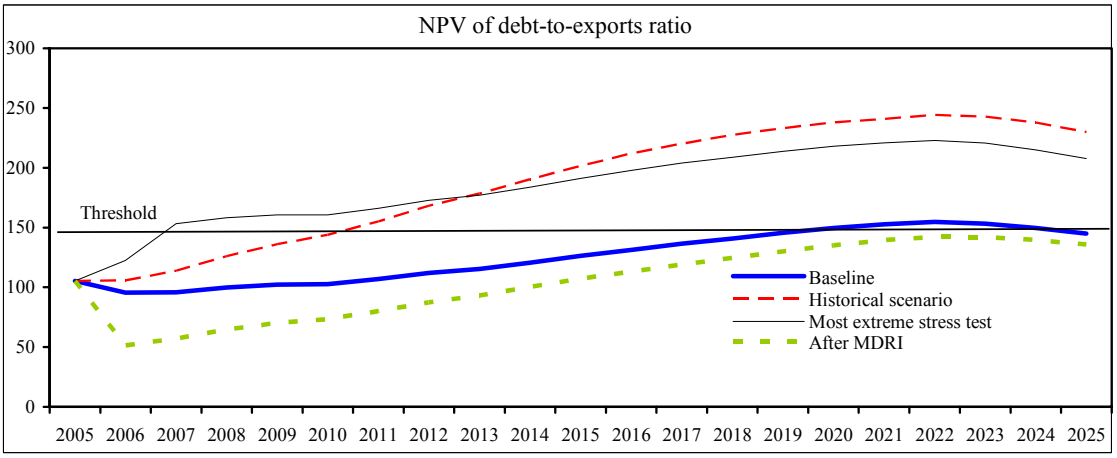
10. **Debt sustainability was also assessed under a Multilateral Debt Relief Initiative (MDRI) scenario**. The MDRI scenario assumes a cut-off date of January 1, 2005 for IMF and African Development Fund debt, and January 1, 2004 for IDA (International Development Association) loans. The scenario assumes an implementation date of January 1, 2006 for IMF and AfDB and July 1, 2006 for IDA. The projected loan disbursements from multilaterals are assumed to be somewhat lowered relative to the baseline as a result of MDRI, reflecting the amount and distribution of compensatory financing for IDA and the AfDB relative to debt service relief. Nonetheless, net transfers from multilaterals would increase as a result of MDRI over the medium term. This scenario results in a halving of the debt-to-GDP ratio in 2006 and reduces the debt service ratio by half over the medium term to a range of 3-5 percent of exports. However, over the longer term (twenty years) the MDRI scenario debt indicators approach those in the baseline scenario (determined by new borrowing over the period 2005-25). The risk of debt distress in the MDRI scenario is low over the medium term, but in the long term would return towards the moderate risk level assessed in the baseline scenario. The scenario is based on current assumptions and may need to be revised when the institution-specific terms of MDRI are finalized.

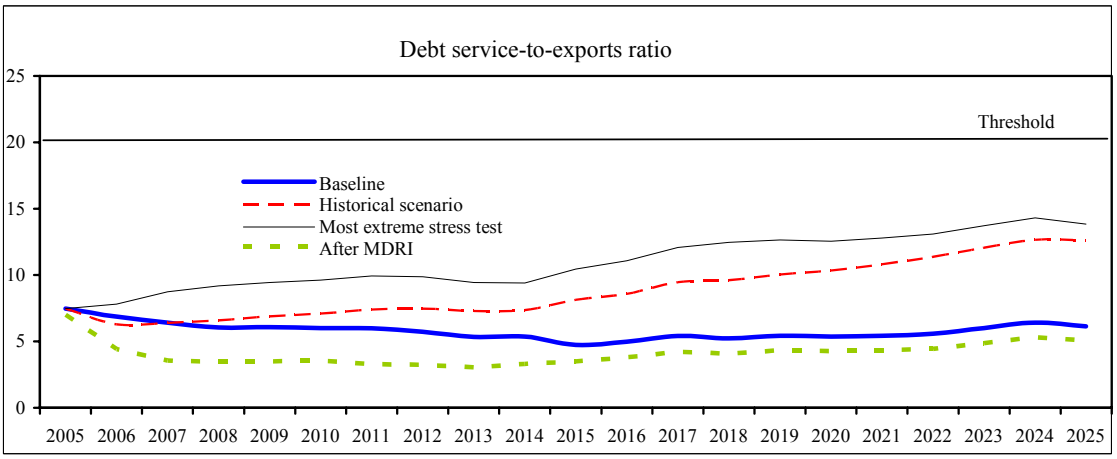
D. Conclusion

11. The analysis indicates that external debt sustainability is broadly achieved under baseline projections in relation to the policy dependent debt burden thresholds. However, adverse external and internal shocks could result in breaching debt-to-exports and debt-to-GDP thresholds over the medium term. Accordingly Mali is assessed as a having a moderate risk of debt distress. Accelerating export growth and maintaining highly concessional loan assistance are important factors in maintaining debt sustainability. The multilateral debt reduction initiative (MDRI) expected to be implemented in 2006 is also projected to improve the debt ratios significantly over the medium term.

Figure 1. Mali: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2005-2025
(In percent)







Source: Staff projections and simulations.

Table 1. Mali: External Debt Sustainability Framework, Baseline Scenario, 2002-2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical	Standard	Estimate	Projections								
	2002	2003	2004	Average 6/	Deviation 6/	2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025	2011-25 Average
External debt (nominal) 1/	**90.2**	**64.0**	**63.1**	91.1	16.6	**67.5**	**66.5**	**66.1**	**66.0**	**65.6**	**65.0**	**66.1**	**61.2**	**39.4**	**55.9**
Of which : public and publicly guaranteed (PPG)	90.2	64.0	63.1	90.3	16.3	67.5	66.5	66.1	66.0	65.6	65.0	66.1	61.2	39.4	55.9
Change in external debt	1.5	-26.3	-0.9	-5.3	10.8	4.4	-0.9	-0.4	-0.1	-0.3	-0.7	0.3	-0.8	-3.7	-1.7
Identified net debt-creating flows	-11.6	-16.2	-2.2	5.1	15.7	6.7	5.3	3.5	3.4	3.1	2.9	4.1	4.5	9.3	6.0
Non-interest current account deficit	**3.4**	**8.2**	**8.2**	**14.2**	**5.5**	**10.9**	**8.8**	**7.4**	**7.1**	**6.8**	**6.7**	**7.9**	**8.3**	**11.8**	**9.3**
Deficit in balance of goods and services	0.2	7.2	7.0	11.3	8.6	10.0	7.8	5.9	5.3	5.0	4.9	6.5	7.5	12.6	9.0
Exports	31.9	26.1	23.2	24.0	4.3	25.9	28.7	29.0	28.2	27.8	27.6	27.9	23.2	15.2	21.0
Imports	32.0	33.3	30.2	35.2	8.6	35.8	36.5	34.9	33.6	32.8	32.5	34.3	30.7	27.8	29.9
Net current transfers (negative = inflow)	-3.6	-3.6	-3.2	-5.2	1.6	-4.1	-3.3	-2.0	-1.9	-1.9	-1.8	-2.5	-1.6	-1.3	-1.5
Other current account flows (negative = net inflow)	6.8	4.6	4.4	8.1	6.9	5.0	4.3	3.5	3.7	3.7	3.6	4.0	2.4	0.4	1.9
Net FDI (negative = inflow)	**-7.2**	**-3.0**	**-1.1**	**-2.3**	**2.0**	**-1.2**	**-0.7**	**-0.8**	**-0.8**	**-0.8**	**-0.8**	**-0.9**	**-0.6**	**-0.2**	**-0.5**
Endogenous debt dynamics 2/	**-7.7**	**-21.4**	**-9.3**	-6.8	11.6	**-3.0**	**-2.8**	**-3.1**	**-2.8**	**-2.9**	**-3.0**	**-2.9**	**-3.2**	**-2.3**	**-2.9**
Contribution from nominal interest rate	0.9	0.5	0.7	0.6	0.3	0.6	0.7	0.6	0.5	0.5	0.5	0.6	0.3	0.1	0.3
Contribution from real GDP growth	-3.4	-4.9	-1.2	-4.8	4.0	-3.5	-3.5	-3.7	-3.3	-3.4	-3.5	-3.5	-3.5	-2.4	-3.2
Contribution from price and exchange rate changes	-5.2	-17.1	-8.8	-2.6	10.3	…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**13.1**	**-10.1**	**1.3**	-10.5	11.2	**-2.3**	**-6.3**	**-3.9**	**-3.6**	**-3.4**	**-3.5**	**-3.8**	**-5.3**	**-13.0**	**-7.7**
Of which : exceptional financing	-1.0	-0.8	-0.6	-2.3	3.7	-0.7	-0.6	-0.5	-0.5	-0.4	-0.4	-0.5	-0.3	-0.1	-0.2
NPV of external debt 4/	...	...	24.7	24.7		27.2	27.5	27.8	28.2	28.4	28.3	27.9	29.3	22.1	28.0
In percent of exports	...	...	106.4	106.4		105.2	95.6	95.8	99.9	102.3	102.8	100.3	126.3	144.9	136.0
NPV of PPG external debt	**...**	**...**	**24.7**	24.7		**27.2**	**27.5**	**27.8**	**28.2**	**28.4**	**28.3**	**27.9**	**29.3**	**22.1**	**28.0**
In percent of exports	...	...	106.4	106.4		105.2	95.6	95.8	99.9	102.3	102.8	100.3	126.3	144.9	136.0
Debt service-to-exports ratio (in percent)	6.3	5.8	6.4	14.2	8.9	7.4	6.8	6.4	6.0	6.1	6.0	6.5	4.7	6.1	5.5
PPG debt service-to-exports ratio (in percent)	6.3	5.8	6.4	13.7	9.1	7.4	6.8	6.4	6.0	6.1	6.0	6.5	4.7	6.1	5.5
Total gross financing need (billions of U.S. dollars)	-0.1	0.3	0.5	0.4	0.2	0.6	0.5	0.5	0.5	0.5	0.6	0.5	1.0	3.2	1.6
Non-interest current account deficit that stabilizes debt ratio	1.8	34.5	9.1	19.5	10.8	6.5	9.7	7.8	7.2	7.2	7.3	7.6	9.1	15.5	11.0
Key macroeconomic assumptions															
Real GDP growth (in percent)	4.3	7.2	2.3	5.5	4.2	5.4	5.4	6.1	5.4	5.6	5.7	5.6	6.1	6.1	6.0
GDP deflator in US dollar terms (change in percent)	6.2	23.3	15.9	4.3	11.9	-7.7	-0.1	3.4	3.2	2.7	2.3	0.6	2.2	2.7	2.5
Effective interest rate (percent) 5/	1.2	0.8	1.3	0.7	0.4	0.9	1.1	1.0	0.8	0.8	0.8	0.9	0.5	0.3	0.5
Growth of exports of G&S (US dollar terms, in percent)	21.7	8.2	5.6	12.4	13.8	8.3	17.1	10.9	5.8	6.7	7.4	9.4	4.5	3.8	4.4
Growth of imports of G&S (US dollar terms, in percent)	-15.0	37.2	7.6	8.8	22.3	15.4	7.4	5.1	4.5	6.0	7.1	7.6	7.5	7.9	7.5
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	52.1	52.1	52.2	52.3	52.3	52.3	52.2	52.3	58.2	53.1
Memorandym item:															
Nominal GDP (billions of US dollars)	3.3	4.4	5.2	3.3	0.9	5.1	5.4	5.9	6.4	7.0	7.5	6.2	11.2	25.9	15.4

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Mali: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-25
(In percent)

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	**2015**	2025
NPV of debt-to-GDP ratio								
Baseline	27	27	28	28	28	28	29	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	27	30	33	36	38	40	**47**	35
A2. New public sector loans on less favorable terms in 2006-25 2/	27	28	29	30	31	31	**34**	28
A3. MDRI Relief from 2006	27	15	17	18	20	20	25	21
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	27	29	30	31	31	31	**32**	24
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	27	30	33	33	33	33	**33**	24
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	27	30	34	34	34	34	**35**	27
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	27	28	28	28	28	28	**29**	22
B5. Combination of B1-B4 using one-half standard deviation shocks	27	29	33	34	34	34	**34**	25
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	27	39	40	40	41	41	**42**	32
NPV of debt-to-exports ratio								
Baseline	105	96	96	100	102	103	**126**	145
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	105	106	114	126	136	144	**202**	230
A2. New public sector loans on less favorable terms in 2006-25 2/	105	97	100	106	111	113	**147**	181
A3. MDRI Relief from 2006	105	51	57	64	70	73	107	136
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	105	96	96	100	102	103	**126**	145
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	105	122	153	158	161	161	**191**	208
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	105	96	96	100	102	103	**126**	145
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	105	96	95	99	101	102	**125**	144
B5. Combination of B1-B4 using one-half standard deviation shocks	105	110	120	124	127	127	**154**	173
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	105	96	96	100	102	103	**126**	145
Debt service ratio								
Baseline	7	7	6	6	6	6	5	6
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2006-25 1/	7	6	6	7	7	7	**8**	13
A2. New public sector loans on less favorable terms in 2006-25 2/	7	7	6	7	7	8	**8**	12
A3. MDRI Relief from 2006	7	4	4	3	4	4	3	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	7	7	6	6	7	7	**8**	10
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	7	8	9	9	9	10	**10**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	7	7	6	6	7	7	**8**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	7	7	6	6	7	7	**7**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	7	7	8	8	8	8	**9**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	7	7	6	6	7	7	**8**	10
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	53	53	53	53	53	53	**53**	53

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assumi an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.